

DeepVision: Heads-up Computing and AI in Education

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ABSTRACT

Heads-up computing together with AI can enhance in-class learning experiences. In this position paper, we propose the development of a multimodal AI system called DeepVision that integrates Automatic Speech Recognition (ASR), Large Language Models (LLM), Large Vision Models (LVM), Information Retrieval (IR) and Inclusive User Experience Design (IUX) to convert real-time lectures into multiple knowledge representations. These will be visualized on heads-up communication devices such as Augmented Reality (AR) and Mixed Reality (MR) devices. The initiative is a collaboration between Habitat Learn Limited (HLL) and the University of Southampton, leveraging HLL's existing software and extensive data repository to address the challenges of traditional and digital learning environments, especially for students with disabilities or language differences.

CCS CONCEPTS

- Human-centered computing → Human computer interaction (HCI) → Interaction paradigms → Mixed / augmented reality;
- Information systems → Language models; Multimedia and multimodal retrieval; Users and interactive retrieval.

KEYWORDS

Heads-up Computing; Large Language Model; AI; AR; Multimodal Information Access and Retrieval; Inclusive User Experience Design

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1 Introduction

Teachers and students have been utilising personal computer or mobile devices to setup e-learning environments to improve teaching and learning experience and effectiveness [10,11,20]. While these tools have greatly improved the learning process after class, students and teachers still have difficulties to adjust teaching and learning routines to accommodate these tools, which require users to adopt a heads-down position instead of focusing on the lecturer presentation [15].

Digital learning is challenging for students with disabilities or language differences. The latest Generative AI together with Computer Vision and Automatic Speech recognition can analyse large amounts of digital learning content and adaptively help students with different learning requirements [18].

Post-pandemic 33% of university courses are online [1]. Citrix reports 79% of students believe increased online learning will negatively impact their studies, requiring better 'scaffolded support' to engage effectively with learning content [4]. Identified shortcomings in blended learning delivery at some HEIs post-pandemic is concerning [16]. Within this trend, more students with disabilities find traditional teaching methods less effective. We propose a heads-up computing system that seeks to address this problem using multimodal AI, which offers alternative ways to engage with content, such as speech-to-text for those with hearing impairments or visual content for those with dyslexia [3].

Heads-up computing [17] is a new interaction paradigm designed to provide seamless support for daily human activities. This user-centric approach alleviates common limitations of current heads-down technologies, which provides unimodal interaction between the device and the user. The heads-up computing employs multimodal interaction such as voice input, gestures image and gaze capture to advance the existing inclusive user-centered information access [6,7,8,12,13]. The development of generative AI further enhances the research and production of the new technologies [2,19]. Especially in education markets, the exiting education technologies will be enriched by the heads-up computing and AI. DeepVision will explore the challenges and opportunities that Heads-up computing will bring to the theories, models and evaluation methods and production of the inclusive and user-centered education technologies.

processing AI algorithms. Such equipment is often cost-prohibitive for schools and higher education institutions (HEIs) and technically challenging to set up and maintain.

3 Proposed Solution

Application of multimodal AI technologies in real-time lectures using AI to output a user-friendly format for heads-up devices is a major challenge as ASR, Large Language Model (LLM), and Large Vision Model (LVM) need to integrate to turn audio-visual lecture content into representations that work for real-time heads-up communication via wearable devices, such as Augmentative or Mixed Reality Glasses.

We proposed a system called DeepVision to combine AI and heads-up computing into an educational environment. Our primary objectives involve researching the application of multimodal AI, including ASR, LLM, and LVM, to create multiple representations of lecture content. This will enable multimodal communication and information access. The project plans to implement these representations into hardware or classroom devices to improve learning outcomes. By combining these representations interactively, DeepVision aims to achieve more inclusive and accessible real-time heads-up interaction and information access during lectures.

To achieve these goals, DeepVision will train algorithms to generate multimodal Knowledge Representations (KRs) using the 450,000 hours of lecture content available in Habitat Learn's repository. These KRs will be visualized on multiple heads-up enabled devices, especially benefiting students with special learning requirements. The visualizations will include automatic captioning using ASR, automatic image and video description using visual language models, interactive key phrases and mind maps, and text descriptions for visual objects (Figure 1).

DeepVision is structured into several key components, each with distinct objectives and tasks designed to collaboratively build a comprehensive heads-up multimodal AI system to enhance educational outcomes.

The first major component, data collection, focuses on designing the Multimodal information access model and data labelling. This phase involves defining the multimodal data transformation processes, such as converting video to transcripts and then to summary notes, and converting images to descriptions. Key tasks include selecting and cleaning up the top lecture recordings from their repository, using semi-automated methods to extract key information necessary for visualization in later stages. This component aims to lay the foundational data and rules that will support the effective functioning of the AI models developed in subsequent phases.

The fine-tuning component is dedicated to train Multimodal AI models, including LLM and LVM. This component is pivotal as it deals with extracting and transforming multimodal data from the labelled documents prepared in the data collection component.

Tasks include developing algorithms to automatically extract relevant data from each document, such as concepts, relations in lecture summaries, and dynamic mind maps, transforming multimedia data into descriptive text and extract the content of the data to enable content-based information retrieval and access. This component also encompasses the development of document interlinking for more effective information retrieval and recommendation to enhance the pedagogical impact of the learning materials.

The final major component, data visualisation and information presentation, revolves around implementing user interfaces (UI) for the different multimodal AI outputs generated in the fine-tuning component. This includes creating mockups and high-fidelity demonstrations for various AI outputs tailored to the hardware being used, such as AR or MR glasses or other heads-up displays. This component is crucial for ensuring that the multimodal AI's outputs are accessible and practically useful in real-time educational settings, thereby enhancing the overall learning experience.

4 Conclusion

The DeepVision system exemplifies a groundbreaking approach to enhancing educational experiences through the integration of heads-up computing, AI and multimodal information access technologies. By systematically designing, developing, and implementing these technologies across three major components, DeepVision aims to transform lecture content and information into interactive, accessible multimodal formats. These include AI-driven summarizations, image and video descriptions, interactive mind maps, and multimodal information access, all visualized through innovative user interfaces tailored for heads-up displays like AR or MR glasses. The project focuses on creating a seamless integration of automatic speech recognition, LLMs and LVMs, and promises to significantly improve the inclusive multimodal information access experiences and learning outcomes for students, particularly those with special learning needs.

Furthermore, the structured implementation of these components, from data labelling and AI model training to the final user interface designs, ensures that the technology is both robust and applicable in real-time educational settings. DeepVision's methodical approach, aiming to enhance how educational content and information is delivered and engaged with, could set new standards in the use of AI in education. By addressing the challenges of accessibility and engagement in learning through cutting-edge technology, DeepVision stands to make a substantial impact on the educational landscape, potentially reducing barriers for students with disabilities and catering to a diverse range of learning preferences.

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